

MOSFET Output Protection Test – Test Summary Report

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Date: 6/9/2025

Boards Tested: 5 prototype SNOW CM boards

Outputs Under Test: 14 per board

FET Part Number: VN9008AJTR

Objective

The objective of this test was to evaluate the new self-protection MOSFETs under unusual electrical conditions. Specifically, the test aimed to determine whether the MOSFETs could survive both double loading and direct shorts to Ground and Power (PWR) without sustaining permanent damage.

Test Setup

- **Custom Test Fixture:**

A custom fixture was designed and built to test across all 14 outputs. The fixture used 28 HYD01638 [solenoid coils](#), with two solenoids wired, in parallel, per output to simulate a double-load condition.

- **Visual Output Indicator:**

An indicator LED was installed on each output line to confirm whether the output was active during testing.

- **Test Boards:**

Five control boards were tested, each with 14 high-side MOSFET-driven outputs.

- **Measurement equipment:**

Fluke 179 True RMS Multimeter

Used for measuring amperage – Certified 09/19/24.

Extech Multimeter

Used for measuring Voltage - Certified 11/27/23.

BK Precision 1688B Power Supply

Used for inputting Voltage – Certified N/A

Keysight DSOX1102G Oscilloscope

Used for looking at the rising and falling edges or unusual behaviors - Certified N/A

Test Procedure

1. Set power supply to 15VDC

- a. (measured 15.07VDC)

2. Initial Functionality Check:

Verified normal operation with all 14 outputs activated in parallel pairs (double-loaded) using the test fixture. LED indicators and coil activation were observed.

3. Short to Ground Test:

Each output was shorted directly to ground while active to simulate a fault condition. The system was monitored for:

- a. FET survival
- b. Whether the output shut down
- c. Recovery after power cycle

4. Short to Power (PWR) Test:

Each output was also shorted to the power to simulate reverse connection.

- a. FET survival
- b. Output turns off with short removed
- c. Normal operation with short removed.

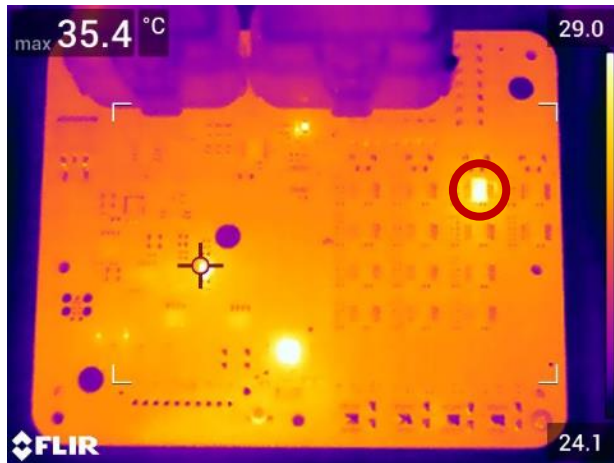
5. Post-Fault Verification:

After each short test, the board was power-cycled and outputs were re-checked using the test fixture to confirm functionality (LED ON, solenoid activation).

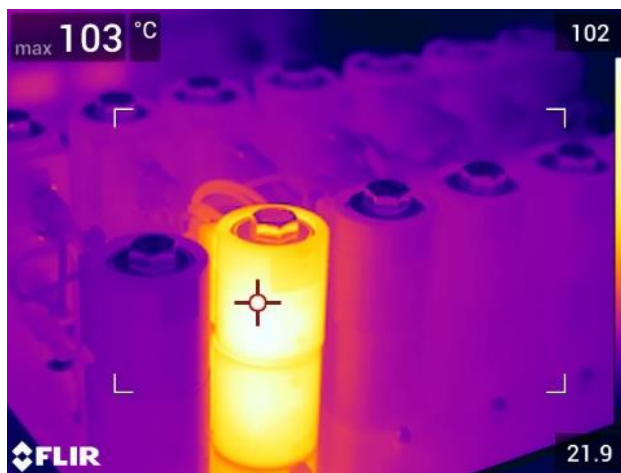
Update – June 20, 2025

- Boards returned from SMT.
- One connector was found sideways in the potting.
 - Potting material and connector were removed.
 - Connector replaced and secured with silicone.
- All pin and short tests were re-run.
 - All tests passed.
- Boards are being sent to Scott at Toro for final testing and confirmation.

Pictures



This is the picture of one of the MOSFET under load. The MOSFET is in the upright of this photo.



This is a picture of the Coils under load.

